

Forest Biomass Solutions

Vance Russell, Joshua Harrison

Takeaways

- The issue at hand is whether active forest management is ecologically legitimate under novel climate conditions. We believe that ecological forest management following the guidelines outlined in GTR-220, coupled with prescribed fire maintenance, will create more resilient forests and reduce fire risk (North *et al.*, 2009).
- Biomass utilization is a downstream process of forest health treatments. It is not a driver of logging. In mitigating wildfire, we need to make the distinction between 1. sawtimber, 2. unavoidable treatment residues, and 3. excess forest biomass streams that exist whether or not there's an energy market.
- Continued business as usual will mean that California's forests continue being a net emitter of carbon. Biomass utilization is not free of greenhouse gas emissions; however, it is part of a solution to greatly reduce massive emissions from uncontrolled wildfires.
- Economics
- Policies that focus on reducing emissions from biomass while integrating forest health solutions with climate change are critical to the long-term biodiversity and human health throughout California and the West.

Background

In December 2025, Shaye Wolf published a [commentary](#) claiming biomass is a money pit that won't solve California's energy or wildfire problems. The piece cited research from authors whose work has been discredited or debunked by a majority of climate and forest scientists. It also made several claims that were neither cited nor accurate. Shortly afterward, Matt Dias of the California Forestry Association responded, arguing that biomass utilization isn't primarily about renewable energy production—rather, the real issue is confronting the wildfire crisis. We now have a perspective from an environmental group and a response from industry.

This article examines the need for forest health treatments that produce biomass and debunks common myths around its utilization. We then proceed to broader wildfire disinformation¹, discuss impacts of forest health on biodiversity, and provide broad solutions.

Biomass

Many biomass critiques rely on broader wildfire myths, e.g., “fires are mostly shrubland,” “thinning increases fire severity,” “forests should be left alone”, or “thinning is an excuse for logging”. Disinformation in this realm becomes a mechanism by which forest management and biomass are broadly maligned without nuance. Because of this, we get a singular argument along the lines of misunderstood fire ecology -> misdiagnosed forest conditions -> misframed biomass debate -> bad policy outcomes. This result does nothing to mitigate the wildfire problem in the West.

THE disagreement

In reality, the debate isn't really about biomass. It's about whether active forest management is ecologically legitimate under novel climate conditions. CBD's position implicitly says restraint and retreat are the answer; yours says intervention, done carefully, is unavoidable. Surfacing that philosophical split would make the argument clearer and more honest.

Forest residuals

¹Ironically, some environmentalists are using a Merchants of Doubt type of approach to their wildfire disinformation campaigns. See Oreskes & Conway (2011) for more on disinformation campaigns intended to sow doubt about science that have been used historically by the petrochemical and cigarette industries, among others.

Biomass utilization is a critical downstream process of forest health treatments. It is not a driver of logging. In mitigating wildfire, we need to make the distinction between 1. sawtimber, 2. unavoidable treatment residues, and 3. Excess forest biomass or residual streams exist whether or not there's an energy market. We also need to make it clear that sawtimber, clearcutting, or excessive forest harvesting cannot be conflated with forest health or wildfire mitigation.

Burning & air quality

Let's acknowledge that older biomass plants were dirty, but modern regulations, controls, and siting are helping to mitigate that issue. These and environmental justice impacts from siting facilities in disadvantaged communities can be further strengthened with strong CEQA and other environmental policies. Nevertheless poor air quality and greenhouse gas emissions from wildfires dwarf regulated point sources. The nearly million acre Dixie fire, California's second largest wildfire, emitted 37 million tonnes of CO_{2e} or the equivalent emissions from electricity consumption of nearly 5 million households (Baldassare, 2024). In 2020 wildfires emitted 127 million tonnes of CO_{2e} nearly two times higher than California's total greenhouse gas emissions reductions achieved since 2003 (Jerrett *et al.*, 2022).

Wolf suggests leaving biomass residues in the forest. Masticating on site, lopping and piling, or removing residue to landings has long been practiced at thinning sites. But the accumulation of materials is overwhelming. A study showed that there are approximately 150,000 acres of piles representing 1,000,000 tonnes with 78% of these materials on public lands (Darlington *et al.*, 2023).² Even disposing of those piles by burning is an episodic, unregulated process that would be much more harmful than utilization in long-term wood products or burning at a facility.

Energy vs. material flow

We need to separate "biomass as energy" from "biomass as material flow management". Wolf collapses everything into burning trees for electricity. Biomass utilization is neither of these an energy silver bullet nor an unending consumer of timber and spewer of wood smoke from giant burning plants. Biomass utilization is one of many solutions to California's wildfire crisis, as the century-long suppression of fires has created unnatural conditions in most forests, making them more susceptible to disease, fire, and other threats related to climate change. We argue for a broader solution: managing unavoidable residues across multiple uses (materials, combined heat and power, heat, and biochar) as part of a landscape risk reduction system.

Economics

Increased forest management, through the use of either hand/mechanical treatments or prescribed fire, can reduce fire suppression costs relative to recent practices by more than US\$400,000 per year (Holland *et al.*, 2022).

Figure 1 shows the cost-benefit of treatments, with an estimated cost of reaching the 1 million acres treated/year goal of 3 billion and a benefit of 7.9/yr (Brown, 2024). A maximized net benefit 22.2 billion/yr would be reached if 3.9 million acres/yr are treated at a cost of 10.5. These numbers seem astronomical, but indicate the sheer scale of the problem of truly reducing wildfire risk after a century of fire suppression. Even if the state reaches the million acre annual treatment goal, all of the biomass produced has to go somewhere, as piling, masticating, and spreading it in place will only shift wildfire risk or destroy the forest understory. That somewhere can be a mix of wood utilization that maximizes carbon sequestration benefits, such as mass timber.

²Only 19% of this material is accessible via existing road networks.

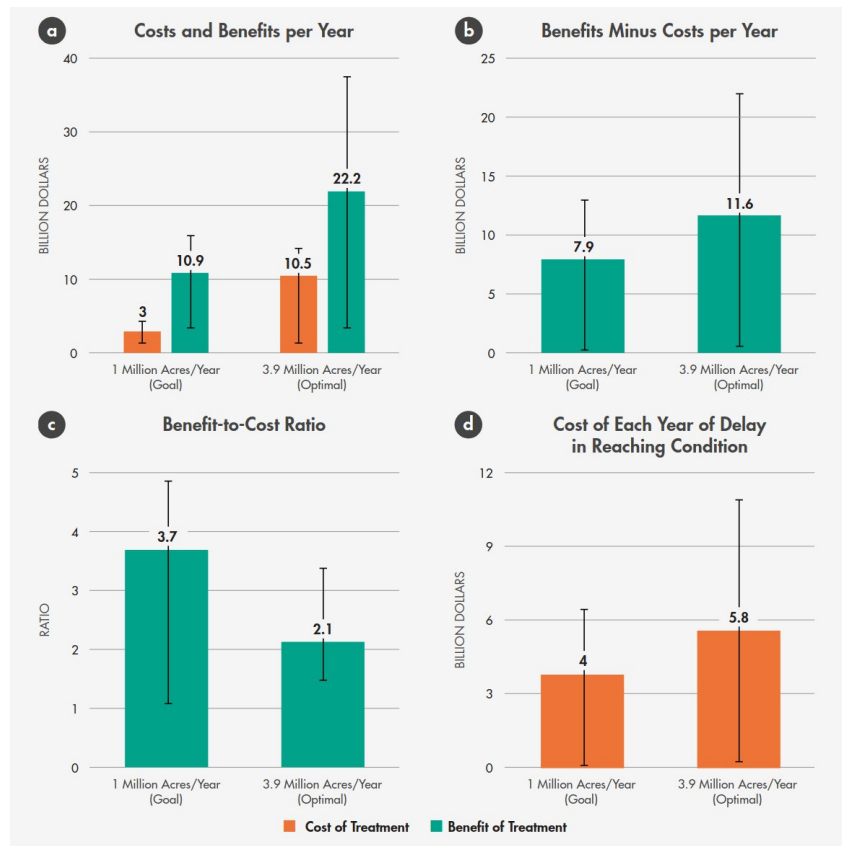


Figure 1: Costs/benefits of California fuel reduction at the goal of 1 million acres vs. 3.9 million acres—the rate that maximizes net benefits (Brown, 2024).

Additional studies estimate a return on investment of treatments of \$3.42 for every \$1 invested (Strabo *et al.*, 2025) similar and a benefit-cost ratio of 4 (Turner, 2023).

BIOMASS FAQs

In addition to wildfire and climate disinformation, several recurring themes touted by some environmentalists deserve debunking and nuance.

1. **Downstream process.** Wolf mentions leaving thinned biomass in place rather than using it as biomass. However, this practice has led to suppression of important undergrowth and, in worst-case scenarios, started or exacerbated some of the largest fires in California's history. For example, it's widely acknowledged that the Dixie Fire, which burned more than 1 million acres across the Plumas National Forest, was spread by a log deck and by thinned fuel piles left in place after thinning operations.
2. **Carbon accounting.** Biomass utilization is not zero-carbon, even with newer technologies that reduce carbon emissions. However, in fire-prone forests, the alternative is large wildfires that release immense amounts of carbon in an uncontrolled manner, often negatively impacting communities downwind for weeks to months. Due to fires, California forests have been a net source of carbon emissions since 2015 (Delyser *et al.*, 2025). Even leaving biomass from thinning projects in place is not a solution, as it shifts wildfire risk or creates unnatural conditions from excessive wood chips spread on the forest floor. Another element of carbon accounting is sequestration. With thinned forests, carbon sequestration increases, and large trees grow larger. In a recent simulation, Elias *et al.* (2025) found a treated forest scenario with 6% more carbon despite a 74% decrease in trees/acre.
3. **Feeding the beast.** A fear among opponents of biomass utilization is that feeding plants from pristine forests to support the biomass industry with dubious climate benefits, e.g., as in North Carolina or Europe (Frisch and Uden 2023). However, as the same authors argue, the situation is

very different in the American West, where biomass results from wildfire mitigation treatments and 90% of forest land is owned by the public and small-scale landowners.

4. **Biomass is an energy silver bullet.** Biomass energy will not replace renewable energy sources such as solar and wind. It is meant to be a local utilization process that is far superior to and less polluting than letting forests burn. The real issue at hand is addressing California's wildfire emergency and not biomass vs. other renewables ([Dias 2025](#)).
5. **Policy-based incentives.** The Bioenergy Market Adjusting Tariff Program (BIOMAT), which the CPUC sadly decided not to renew, required utilities to buy power from small local projects that convert forest biomass to energy. Creating a successor program that could get facilities across the line to function or re-idle facilities like Loyaltan that aren't currently operating is critical to processing biomass from forest health projects.

Wildfire disinformation

Similar to climate misinformation, untruths about wildfire, forests, and logging have multiplied in the media, political discourse, and across organizations (Jones *et al.*, 2022). Likewise, overexposure of wildfire and biomass contrarians in the media to give 'balanced' views can result in public confusion and weakened support for interventions, as well as undermine scientific consensus (Cook, 2020). A perfect example from the CalMatters article is the claim that 'Most of California's destructive wildfires...have burned in shrublands and grasslands...' The citation is for a study that concludes that destructive wildfires in the continental US have occurred mostly in these habitats. In reality, the most destructive California wildfires have occurred in a mix of forests and shrublands.

Fire-adapted forests

Another misconception mentioned is that thinning makes cool, moist forests hotter, drier, and more wind-prone, where the author cites Hagmann *et al.* (2021). Several authors cited in this article are described in Peery *et al.* (2019) as engaging in agenda-driven pseudoscience, such as the selective use of data, mixing science and litigation without declaring conflicts of interest, and harassing scientists publishing competing studies. Yet there's widespread scientific consensus that fire was more frequent in western forests (Stephens *et al.*, 2007), trees were bigger and forest stands were more 'park like' (North *et al.*, 2009), and due to loss of moderate fires due to decades of fire suppression make current conditions in nearly all forests more vulnerable to drought and fire, particularly in a warming climate (Hagmann *et al.*, 2021). Current needs assessments indicate more than 11.2 million forested acres require treatment to reduce future wildfire risk and severity (Delyser *et al.*, 2025). The same study shows that rapid initial treatments can restore resilience before climate change intensifies wildfires.

More importantly, recent fires in mixed conifer woodlands, such as the Dixie Caldor Fires in 2021, showed that thinning and prescribed fire reduced the probability of stand-replacing wildfire and fire severity (Shive *et al.*, 2024; Baldassare, 2024). In fact, the Caldor Fire was deflected by a prescribed burn at Caples Lake the previous year, reducing the risk to surrounding forests and communities (Eagleston *et al.*, 2025; Safford & Saberi, 2026).

Fire suppression

Quite possibly the biggest miss of the Calmatter's article is not acknowledging the degree of ecological departure from 'natural' conditions due to a century of wildfire suppression (Collins *et al.*, 2017; Peery *et al.*, 2019). This human-induced forest condition has led to increasingly large, more frequent fires that destroy ecosystems and communities alike. Prichard *et al.* (2021) found that the effects of fire exclusion are not overstated, and that intentional management that adapts to current forest conditions is needed. Utilizing biomass from projects intended to return forest stands to natural fire regimes will be needed in the short term until prescribed fire or fires allowed to burn can allow the forest to catch up to pre-Columbian conditions.

Biodiversity

Curiously, despite the author's organizational mission, biodiversity wasn't mentioned at all in the opinion piece. For instance, it is well known that the indigenous burning as a part of historic fire regimes enhanced biodiversity (Hoffman *et al.*, 2021). Measuring the impact of thinning, prescribed fire, and other forest health projects should be a feature of any publicly funded projects.

The science advisory panel of the Wildfire Task Force has done a superb job ensuring biodiversity and listed species data are available for all regions in the state, but more can be done to measure project impacts on vulnerable or threatened species and overall impacts on species over time. There is some evidence that a combination of thinning and prescribed fire significantly increases plant diversity in mixed or dry conifer forests (Dodson *et al.*, 2008). However, California is very diverse, and what works in one ecosystem may not work in others, especially in chaparral or mesic systems, where burn frequency is generally too high and thinning or prescribed fire are not applicable treatments.



Solutions

Ms. Wolf did get two things right in her op-ed: prioritize thinning in the wildland urban interface and limit development in fire-prone areas. These solutions, however, fail to address the current problem of excessive biomass in forests, resulting from a century of fire suppression.

“The choice is not between profit and responsibility. It is between managed transition and unmanaged failure.” Harrison (2025)

Removing many small-diameter trees followed by prescribed fire will not only reduce wildfire risk in priority areas (note Caldor fire burn deflected by Caples Lake prescribed fire), but it will reduce excessive competition, causing chronic stress across forest stands, reducing disease, insect invasions, allowing large trees to grow larger, and making forest stands more resilient (North *et al.*, 2022). Either/or solutions are not needed; both/and, integrated solutions with all hands on deck to mitigate wildfire threats to forests and communities. Some of these include

1. **Climate change and biomass.** It is clear that rapid scaling of forest health treatments, including thinning and prescribed fire, is critical to resilient forests in the face of climate change (Delyser *et al.*, 2025). Creating policies that integrate treatments and biomass utilization with climate goals and greenhouse gas reduction enables wildfire risk mitigation and incorporates utilization infrastructure regulations into the same solutions.
2. **Mobile modular technology.** Processing biomass is a temporary solution to the buildup of biomass over a century of fire suppression. Creating biomass infrastructure should not create a feed-the-beast situation, as has happened in the Southeast US or parts of Europe. Rather, processing biomass in areas of need may be needed over time, not only to prioritize local utilization, but also to support the transport of biomass and its products.
3. **Pro-active disaster insurance.** It is clear that repeated large-scale wildfires will make California and other regions of the world uninsurable and possibly uninhabitable. Creating economic incentives to treat prevention will reduce losses, stabilize premium costs, and lower overall impacts to the public (Harrison, 2025). Disaster should not form the basis of a growth economy; policy and market incentives can provide carrot-and-stick approaches to ensure change at the scale that climate and resulting wildfires pose.
4. **Jobs in the woods.** Biomass production and processing can create valuable jobs in the woods in rural, often disadvantaged communities that are threatened by wildfire and also experiencing a constant state of brain drain and loss of capacity due to residents leaving to look for jobs elsewhere (when jobs don't exist in their communities). Creating whole system stewards could be a component of this job creation process.

5. **Biodiversity.** Biodiversity isn't a solution to biomass utilization, per se, but it should be included in every conversation about forest health, e.g., what is the impact on biodiversity of thinning, prescribed fire, and biomass utilization? Removing all snags from the forest to reduce fire risk will negatively impact biodiversity. Indiscriminate thinning, instead of creating gaps and clumps, and ecological-based forest management will reduce biodiversity and may not even work to create resilient forest health stands.

Authors

Vance Russell has nearly 40 years of experience working in forest science & management, rewilding, biodiversity conservation, agricultural landscapes, restoration, and natural resources management. He is the owner of 3point.xyz, where he works for various non-profit, state/federal agencies, and private businesses. Vance is the former Board Chair of Groundswell International, is a trustee for the South Downs National Park Trust, and serves on the Rewilding Leadership Council for the Rewilding Institute.

Joshua Harrison is Director of the **Center for the Study of the Force Majeure** at the University of California, Santa Cruz, where he leads interdisciplinary teams designing regenerative systems in response to climate disruption. His work bridges art, science, and policy: from immersive installations on ocean collapse to circular-economy models for wildfire-damaged forests and allyship with tribal groups bringing back cultural burning.

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